

YEAR 7 MATHEMATICS · CHAPTER 9

The number plane

9.01 Grid references on maps**9.03** The number plane**9.05** Tables of values from rules**9.02** Points in the first quadrant**9.04** Plotting tables of values**9.06** Graphing linear equations

Name _____

Class _____

Teacher _____

How to use this booklet

- 1 Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- 2 Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- 3 Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- 4 At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- 5 Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

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In this chapter you will

- use grid references on a map
- plot and read points in the first quadrant
- plot and read points in all four quadrants
- plot points from tables and describe patterns
- complete tables of values from rules
- graph linear equations and test whether a point is on a line

NSW Mathematics K-10 Syllabus (2022), Stage 4 Number and algebra: MA4-LIN-C-01 creates and displays number patterns and graphs linear relationships on the Cartesian plane.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
9.01

Grid references on maps

Date _____

Learning intentions

We are learning to:

- use grid references to find and describe locations on a map

Success criteria

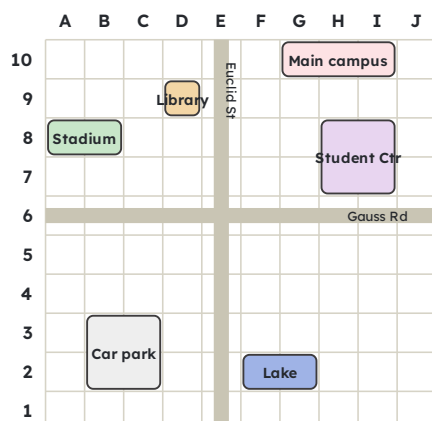
I can:

- ☐ read a grid reference: letter (across) then number (up)
- ☐ state the feature at a grid reference
- ☐ write the grid reference of a feature

Key terms Write the definition as your teacher explains it.**Grid reference****Map**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Use the map in Set A. What is at F2?**Example 2** What is at D9?**YOU DO** On your own: Set A. Use the map. What is at:

① F2

② D9

③ A8

④ H7

YOU DO On your own: Set B. Write a grid reference for:

1 the library

4 the car park

2 the lake

5 the student centre

3 the stadium

6 the main campus

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Write the grid reference of the library.**Example 4** Which grid squares does the car park cover?**YOU DO** On your own: Set C. Use the map.

1 Name a grid square that Gauss Road passes through.

3 Where do Gauss Road and Euclid Street meet?

5 How many squares is the lake from the car park, across?

2 Name a grid square that Euclid Street passes through.

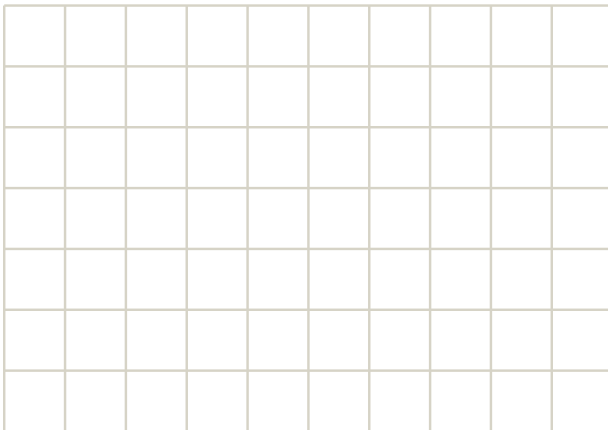
4 Which feature is furthest north (highest row)?

6 Which features are in column H?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Describe how to walk from the lake to the library along the roads.
Example 6 Why is the letter written before the number?
YOU DO On your own: Set D. Draw your own.**1** Draw a map of your school on the grid. Mark 5 places and list their grid references.**2** Write directions from one place to another using grid references.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is at A8?

Level 2

Write the grid reference of the lake.

Level 3

From D9 go 4 right and 2 down. What is there?

Extension · Stretch yourself

Worked example

A treasure is 3 squares right and 4 squares up from B2. Where is it?

- 3 squares right of B is E
- 4 squares up from 2 is 6

Answer: E6

YOU DO

On your own: read the worked example, then try these.

- E1** From H7, go 5 left and 2 down. Where are you?

- E2** How many squares is it from A1 to J10 if you can only move up or right?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A grid reference names a square.
 2. Letter first (across), then number (up).
 3. Large features may cover several squares.
 4. Check by running your finger across, then up.

Level 1

What is at F2?

1. Column F, row 2

Answer: the lake

Level 2

Library

1. Column D, row 9

Answer: D9

Level 3

Car park

1. Covers 4 squares

Answer: B2, C2, B3, C3

Extension

3 right, 4 up from B2

1. B → E, 2 → 6

Answer: E6

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.02

Points in the first quadrant

Date _____

Learning intentions

We are learning to:

- plot and read points on a number plane in the first quadrant

Success criteria

I can:

- ☐ know the **x-coordinate** is across and the **y-coordinate** is up
- ☐ plot a point (x, y)
- ☐ read the coordinates of a point

Key terms Write the definition as your teacher explains it.

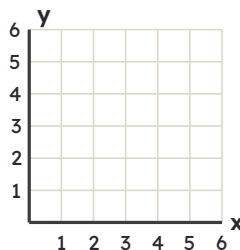
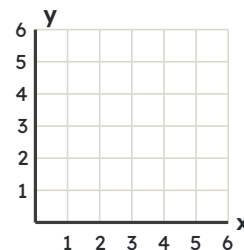
Coordinates

Origin

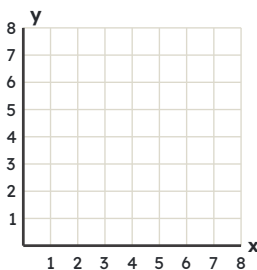
Axis

Notes

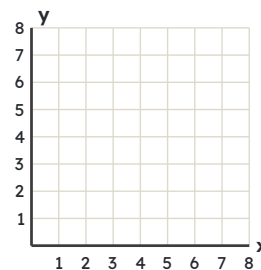
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Plot A(2, 3) and B(1, 6).**Example 2** Plot C(3, 0) and D(0, 4).**YOU DO** On your own: Set A. Plot and label each point on the same number plane.

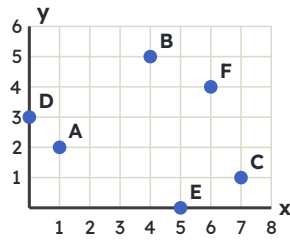
1 A(2, 3), B(1, 6), C(3, 0)



2 D(0, 4), E(5, 2), F(6, 5)



YOU DO On your own: Set B. Write the coordinates of each point.



1 A

2 B

3 C

4 D

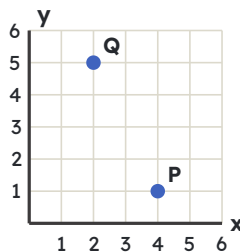
5 E

6 F

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

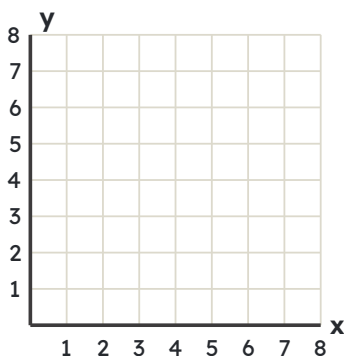
Example 3 Write the coordinates of P and Q.



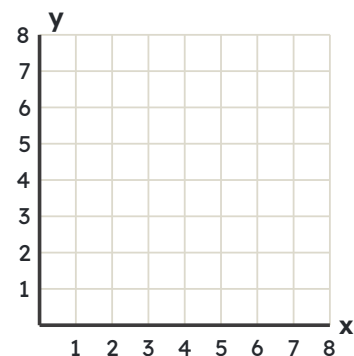
Example 4 Is (2, 5) the same point as (5, 2)? Explain.

YOU DO On your own: Set C. Plot, join in order and name the shape.

1 (1, 1), (6, 1), (6, 4), (1, 4), (1, 1)



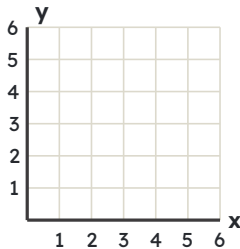
2 (2, 1), (6, 1), (4, 6), (2, 1)



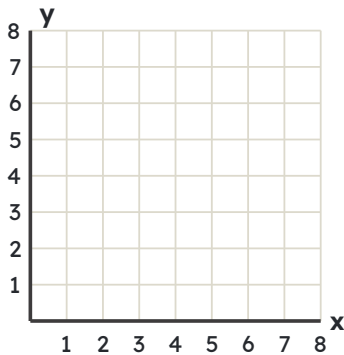


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Plot $(1, 1)$, $(5, 1)$, $(5, 4)$, $(1, 4)$ and join them. Name the shape.**Example 6** What are the coordinates of the origin?
YOU DO On your own: Set D. Think it through.

- 1 Three corners of a square are $(1, 2)$, $(5, 2)$ and $(5, 6)$. Plot them and find the fourth corner.



- 2 Name 3 points on the x -axis and 3 on the y -axis. What do you notice?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1What is the x -coordinate of $(7, 3)$?
Level 2Where is the point $(0, 5)$?
Level 3A rectangle has corners $(1, 2)$, $(6, 2)$, $(6, 5)$. Find the fourth corner.

Extension · Stretch yourself

Worked example

Find the midpoint of (2, 3) and (6, 7).

1. Halfway across: $(2 + 6) \div 2 = 4$
2. Halfway up: $(3 + 7) \div 2 = 5$

Answer: (4, 5)

YOU DO

On your own: read the worked example, then try these.

- E1** Find the midpoint of (1, 8) and (5, 2).

- E2** A rectangle has corners (1, 1) and (7, 5). Find its area in square units.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. (x, y): x across first, then y up.
 3. Points on the x-axis have $y = 0$.

2. Start at the origin (0, 0).
4. Points on the y-axis have $x = 0$.

Level 1

Plot A(2, 3)

1. 2 across, 3 up

Answer: a dot labelled A

Level 2

Coordinates of P

1. 4 across, 1 up

Answer: (4, 1)

Level 3

(2, 5) vs (5, 2)

1. Order matters

Answer: different points

Extension

Midpoint of (2, 3), (6, 7)

1. Average x and y

Answer: (4, 5)

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.03

The number plane

Date _____

Learning intentions

We are learning to:

- plot and read points in all four quadrants

Success criteria

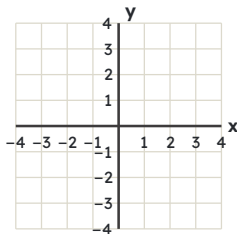
I can:

- ☐ name the four quadrants
- ☐ plot points with negative coordinates
- ☐ state which quadrant a point is in from its signs

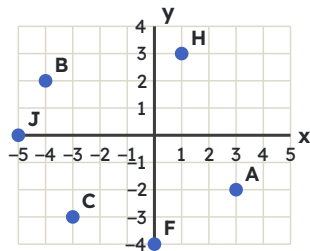
Key terms Write the definition as your teacher explains it.**Quadrant****Origin**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Label the quadrants 1st to 4th.**Example 2** In which quadrant is $(-3, -3)$?**YOU DO** On your own: Set A. Which quadrant or axis is each point on?1 $(-3, -3)$ 4 $(3, -6)$ 2 $(4, 2)$ 5 $(0, 4)$ 3 $(-1, 5)$ 6 $(-2, 0)$

YOU DO On your own: Set B. Write the coordinates of each point.



1 H

2 B

3 C

4 A

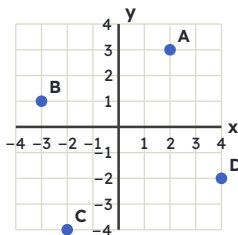
5 F

6 J

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

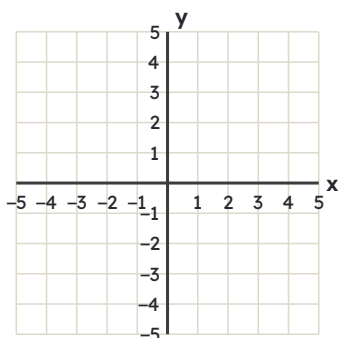
Example 3 Write the coordinates of each point.



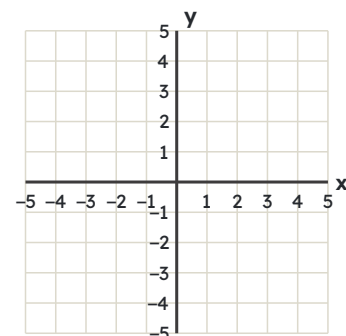
Example 4 Which quadrant has a positive x-coordinate and a negative y-coordinate?

YOU DO On your own: Set C. Plot and join.

- 1 (1, 3), (2, 3), (2, 2), (3, 1), (2, 0), (2, -1), (1, -1), (0, -2), (-1, -1), (-2, -1), (-2, 0), (-3, 1), (-2, 2), (-2, 3), (-1, 3), (0, 4), (1, 3). What shape is it?



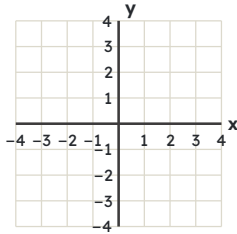
- 2 Plot A(-4, -2), B(2, -2), C(2, 3). Find D so ABCD is a rectangle.





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Plot (1, 3), (2, 2), (3, 1), (2, 0), (1, -1), (0, -2), (-1, -1) and join.**Example 6** Write a point in the 2nd quadrant and a point on the y-axis.**YOU DO** On your own: Set D. Think it through.

- 1 In which quadrant is a point with a negative x-coordinate and a positive y-coordinate? A 1st B 2nd C 3rd D 4th

- 2 What are the coordinates of the origin?

- 3 Reflect (3, -2) in the y-axis. Write the new point.

- 4 What is the distance between (-3, 2) and (4, 2)?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

In which quadrant is (-4, 2)?

Level 2

Write the coordinates of a point in the 4th quadrant.

Level 3

Reflect (-2, 5) in the x-axis.

Extension · Stretch yourself

Worked example

A square has corners $(-2, -1)$ and $(3, -1)$ along its bottom. Find the other two corners (above).

- Side length = $3 - (-2) = 5$
- Go up 5: $y = -1 + 5 = 4$

Answer: $(3, 4)$ and $(-2, 4)$

YOU DO

On your own: read the worked example, then try these.

- E1** A triangle has corners $(-3, 0)$, $(3, 0)$ and $(0, 4)$. Find its area.

- E2** Reflect the triangle $(1, 1)$, $(4, 1)$, $(1, 3)$ in the x-axis. Write the new corners.

Lesson summary

Missed the lesson? Start here.

- Remember**
- The axes cross at the origin $(0, 0)$.
 - 1st $(+, +)$, 2nd $(-, +)$, 3rd $(-, -)$, 4th $(+, -)$.
 - Quadrants go anticlockwise from top right.
 - Points on an axis are in no quadrant.

Level 1

$(-3, -3)$

- Both negative

Answer: 3rd quadrant

Level 2

Point B

- 3 left, 1 up

Answer: $(-3, 1)$

Level 3

$+x, -y$

- Bottom right

Answer: 4th quadrant

Extension

Square on $(-2, -1)$, $(3, -1)$

- Side 5, go up

Answer: $(3, 4)$, $(-2, 4)$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.04

Plotting tables of values

Date _____

Learning intentions

We are learning to:

- plot points from a table and describe the pattern

Success criteria

I can:

- ☐ read (x, y) pairs from a table
- ☐ plot the points and join them with a ruler
- ☐ describe the pattern (straight line, vertical, horizontal, increasing, decreasing)

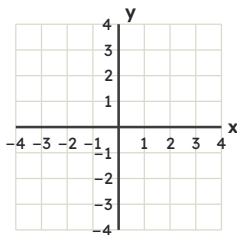
Key terms Write the definition as your teacher explains it.**Table of values****Pattern**

Notes

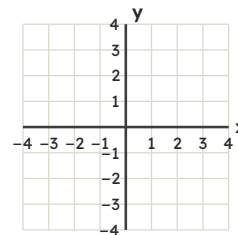
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Plot the table and describe the pattern.

x	-2	-2	-2	-2
y	-2	-1	0	1

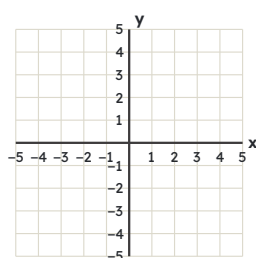
**Example 2** Plot and describe.

x	-2	-1	0	2
y	3	3	3	3

**YOU DO** On your own: Set A. Plot each table and join the points.

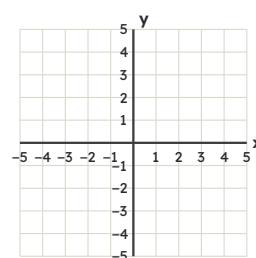
1

x	-2	-2	-2	-2
y	-2	-1	0	1



2

x	-2	-1	0	2
y	3	3	3	3



YOU DO On your own: Set B. Write the table as a list of points.

1

x	0	1	2
y	5	4	3

3

x	1	2	3
y	2	4	6

2

x	-1	0	1
y	-1	1	3

4

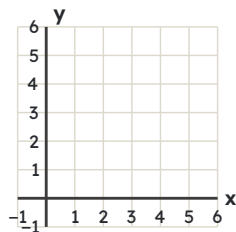
x	0	2	4
y	1	1	1

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

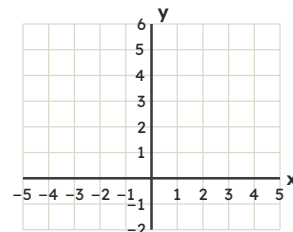
Example 3 Plot and describe.

x	0	1	2	5
y	5	4	3	0



Example 4 Plot and describe.

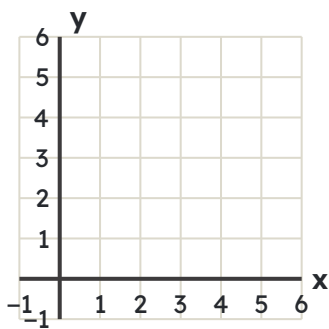
x	-1	0	1	2
y	-1	1	3	5



YOU DO On your own: Set C. Plot, join and describe the pattern.

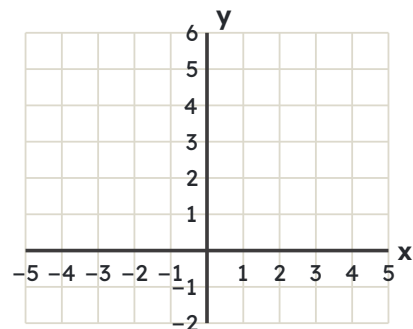
1

x	0	1	2	5
y	5	4	3	0



2

x	-1	0	1	2
y	-1	1	3	5





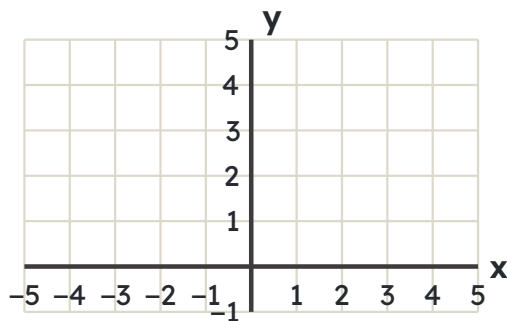
Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** In the table in Example 4, what happens to y when x goes up by 1?
Example 6 Predict y when $x = 3$ in Example 4.
YOU DO On your own: Set D. Plot and describe.

1

x	-2	-1	0	1	2
y	4	1	0	1	4



2 Is the pattern in Question 1 a straight line? Describe it.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Write the points in this table: $x = 0, 1, 2$ and $y = 3, 5, 7$.
Level 2Describe the pattern of the points $(1, 4), (2, 4), (3, 4)$.
Level 3What happens to y each time x goes up by 1 for $(0, 1), (1, 4), (2, 7)$?

Extension · Stretch yourself

Worked example

The points (0, 2), (1, 5), (2, 8) follow a pattern.
Find y when $x = 10$.

- y goes up by 3 each time x goes up by 1
- Start at 2 when $x = 0$: $y = 3x + 2$
- $3 \times 10 + 2$

Answer: 32

YOU DO

On your own: read the worked example, then try these.

- E1** The points (0, 7), (1, 5), (2, 3) follow a pattern.
Find y when $x = 6$.

- E2** Find the rule for the points (1, 4), (2, 8), (3, 12).

Lesson summary

Missed the lesson? Start here.

Remember

- Each column of a table is a point (x, y).
- Plot each point carefully.
- Join them with a ruler if they line up.
- Describe: horizontal, vertical, going up or going down.

Level 1

x always -2

- All points above/below each other

Answer: vertical line

Level 2

y always 3

- All points level

Answer: horizontal line

Level 3

(0, 5), (1, 4), (2, 3)

- y goes down 1

Answer: line going down

Extension

(0, 2), (1, 5), (2, 8)

- $y = 3x + 2$

Answer: 32 when $x = 10$

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LESSON
9.05

Tables of values from rules

Date _____

Learning intentions

We are learning to:

- complete a table of values from a rule

Success criteria

I can:

- ☐ substitute each x value into the rule
- ☐ use negative numbers carefully
- ☐ write the rule from a table

Key terms Write the definition as your teacher explains it.**Rule****Substitute**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Complete for $y = x + 3$.

x	-1	0	1	2	3
y					

Example 2

Complete for $q = 2p - 7$.

p	-2	-1	0	1	2
q					

YOU DO

On your own: Set A. Complete each table.

① $y = x + 3$

x	-1	0	1	2	3
y					

③ $y = x - 4$

x	0	1	2	3	4
y					

② $y = 2x$

x	-1	0	1	2	3
y					

④ $y = 5x$

x	-2	-1	0	1	2
y					

YOU DO On your own: Set B. Complete each table.

① $q = 2p - 7$

p	-2	-1	0	1	2
q					

② $d = 5 - c$

c	-1	0	1	2	3
d					

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Complete for $d = 5 - c$.

c	-1	0	1	2	3
d					

Example 4 Complete for $y = 3x + 1$.

x	-2	-1	0	1	2
y					

YOU DO On your own: Set C. Complete each table.

① $y = 3x + 1$

x	-2	-1	0	1	2
y					

③ $y = \frac{1}{2}x - 2$

x	-4	-2	0	2	4
y					

② $y = -2x + 3$

x	-2	-1	0	1	2
y					

④ $y = 10 - 3x$

x	0	1	2	3	4
y					



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find the rule.

x	0	1	2	3
y	4	6	8	10

Example 6 Find the rule.

x	1	2	3	4
y	7	6	5	4

YOU DO On your own: Set D. Find the rule for each table.

1

x	0	1	2	3
y	4	6	8	10

3

x	0	1	2	3
y	-3	1	5	9

2

x	1	2	3	4
y	7	6	5	4

4

x	-1	0	1	2
y	3	0	-3	-6

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1For $y = x + 5$, find y when $x = -2$.**Level 2**For $y = 4x - 1$, complete for $x = 0, 1, 2$.**Level 3**Find the rule: $x = 0, 1, 2$ and $y = 1, 4, 7$.

Extension · Stretch yourself

Worked example

Complete a table for $y = x^2 - 1$ using $x = -2$ to 2.

1. Square first, then subtract 1
2. $(-2)^2 - 1 = 3$

Answer: $y = 3, 0, -1, 0, 3$

YOU DO

On your own: read the worked example, then try these.

- E1** Complete a table for $y = x^2 + x$ using $x = -2$ to 2.

- E2** Find the rule: $x = 1, 2, 3, 4$ and $y = 1, 4, 9, 16$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write the rule.
 2. Replace x with each value (use brackets for negatives).
 3. Calculate y .
 4. To find a rule: how much does y change when x goes up 1? Then find the start value.

Level 1

$$y = x + 3$$

1. $-1 + 3 = 2, \dots$

Answer: 2, 3, 4, 5, 6

Level 2

$$q = 2p - 7$$

1. $2 \times (-2) - 7 = -11$

Answer: $-11, -9, -7, -5, -3$

Level 3

Rule for 4, 6, 8, 10

1. +2 each time, start 4

Answer: $y = 2x + 4$

Extension

$$y = x^2 - 1$$

1. Square first

Answer: 3, 0, -1, 0, 3

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LESSON
9.06

Graphing linear equations

Date _____

Learning intentions

We are learning to:

- graph a linear equation using a table of values
- test whether a point lies on a line

Success criteria

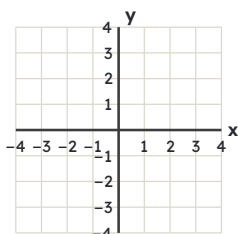
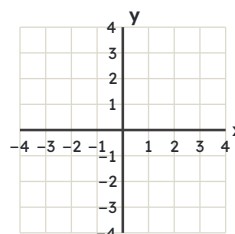
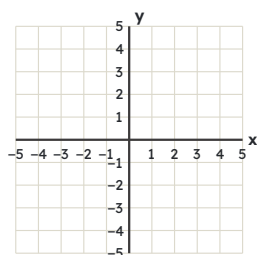
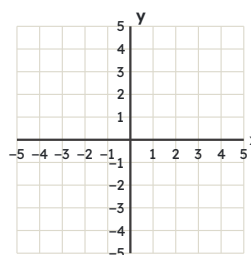
I can:

- ☐ complete a table of values for the equation
- ☐ plot the points and rule a straight line through them
- ☐ substitute a point into the equation to test it

Key terms Write the definition as your teacher explains it.**Linear****Equation**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Graph $y = 2x - 1$ for $x = -1$ to 2.**Example 2** Graph $y = -x + 3$.**YOU DO** On your own: Set A. Complete a table, then graph.**1** $y = x + 1$ ($x = -2$ to 2)**2** $y = 2x - 1$ ($x = -1$ to 2)

YOU DOOn your own: Set B. Does the point lie on $y = 3x + 2$? Write yes or no.

1 (1, 2)

4 (0, 2)

2 (-1, 4)

5 (1, 5)

3 (-2, -4)

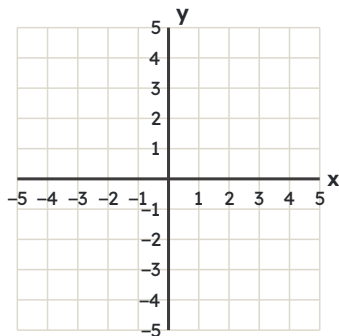
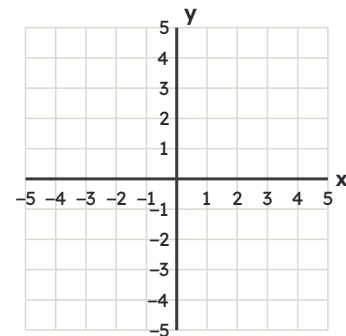
6 (2, 8)

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

Example 3Does (1, 5) lie on $y = 3x + 2$?**Example 4**Does (-2, -4) lie on $y = 3x + 2$?**YOU DO**

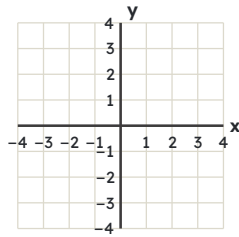
On your own: Set C. Complete a table, then graph.

1 $y = -x + 3$ 2 $y = \frac{1}{2}x - 2$ 

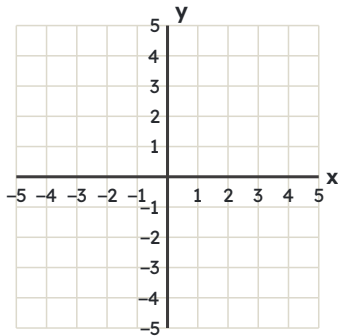


Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Graph $y = \frac{1}{2}x - 2$ for $x = -4$ to 4 .**Example 6** Where does $y = 2x - 1$ cross the y-axis?**YOU DO** On your own: Set D. Graph and compare.

- 1 Graph $y = x$ and $y = x + 2$ on the same plane. What do you notice?



- 2 Where do $y = x + 1$ and $y = -x + 3$ meet? (Use your graphs.)

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1For $y = x + 4$, find y when $x = 1$.

Level 2Does $(2, 5)$ lie on $y = 2x + 1$?

Level 3Does $(-1, 3)$ lie on $y = -2x + 1$?

Extension · Stretch yourself

Worked example

Find the value of k if $(2, k)$ lies on $y = 4x - 3$.

1. Substitute $x = 2$
2. $k = 4 \times 2 - 3$

Answer: $k = 5$

YOU DO

On your own: read the worked example, then try these.

- E1** Find m if $(m, 11)$ lies on $y = 2x + 5$.

- E2** Which of these lines are parallel: $y = 2x + 1$, $y = x + 2$, $y = 2x - 3$?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Make a table with 3 or more x values.
 3. Plot and rule a line through all the points.

2. Substitute to find y .
4. Test a point: substitute x and see if you get its y .

Level 1

$$y = 2x - 1$$

1. $x: -1, 0, 1, 2 \rightarrow y: -3, -1, 1, 3$

Answer: a straight line through $(0, -1)$

Level 2

$$(1, 5) \text{ on } y = 3x + 2?$$

1. $3 \times 1 + 2 = 5 \checkmark$

Answer: yes

Level 3

$$(-2, -4)$$

1. $3 \times (-2) + 2 = -4 \checkmark$

Answer: yes

Extension

$$(2, k) \text{ on } y = 4x - 3$$

1. $8 - 3$

Answer: $k = 5$

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